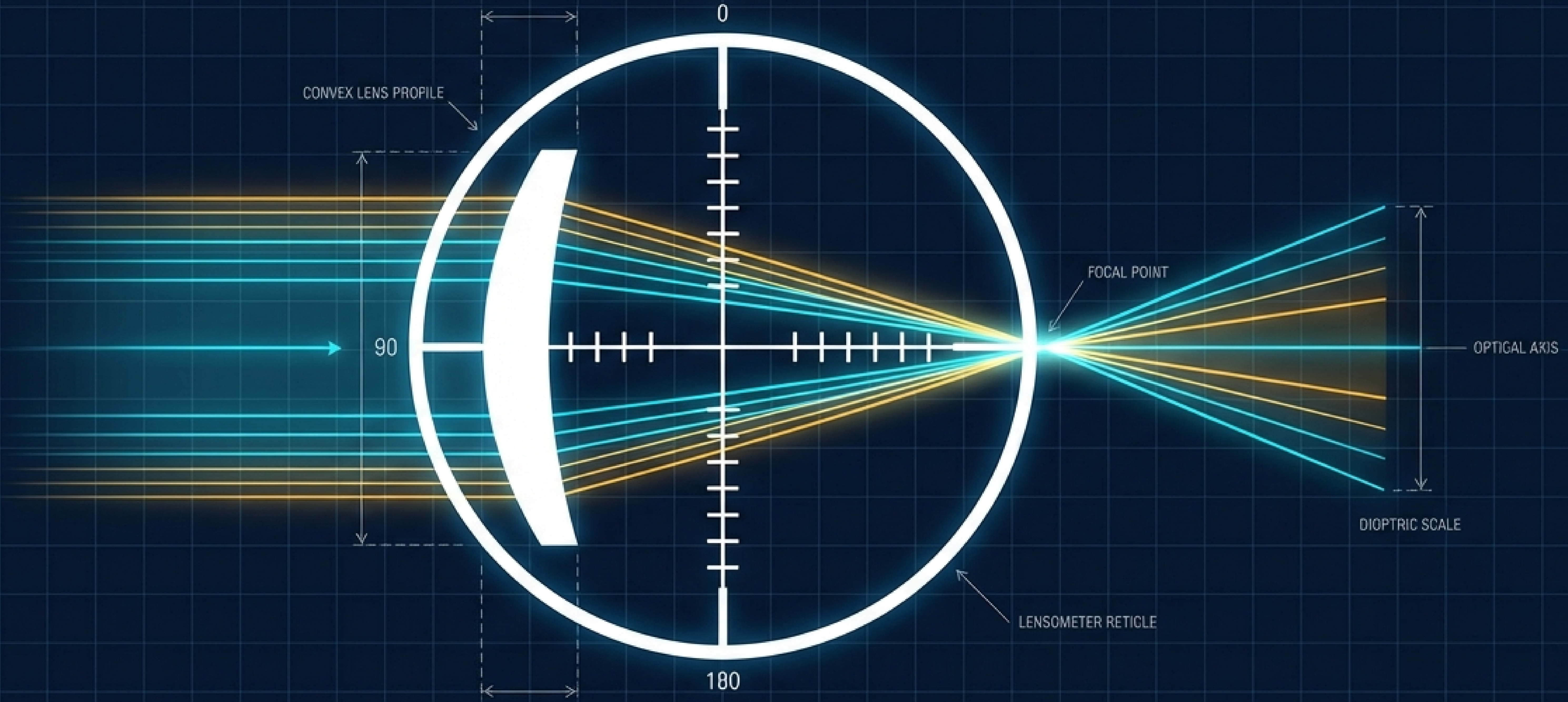
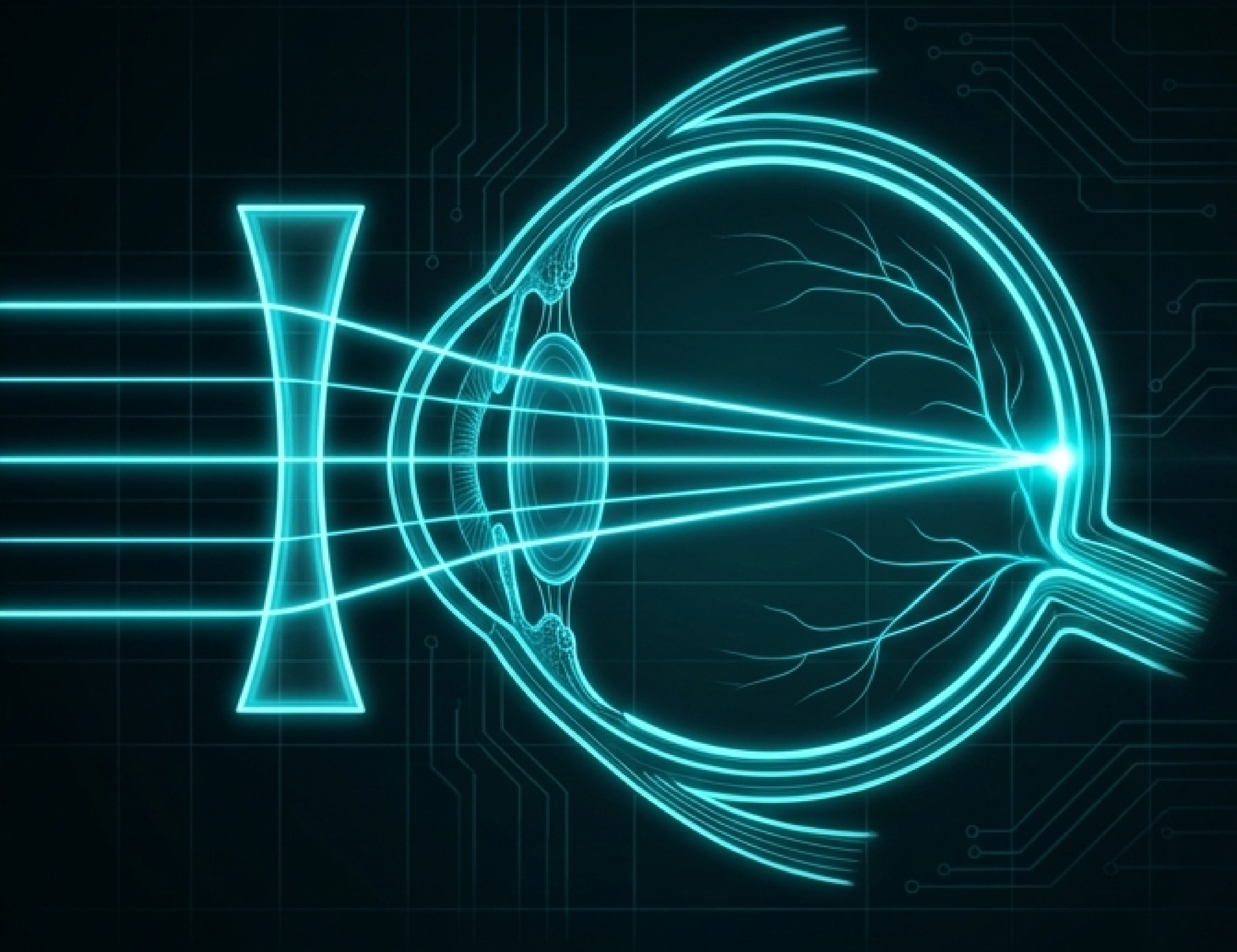
Decoding the Lens

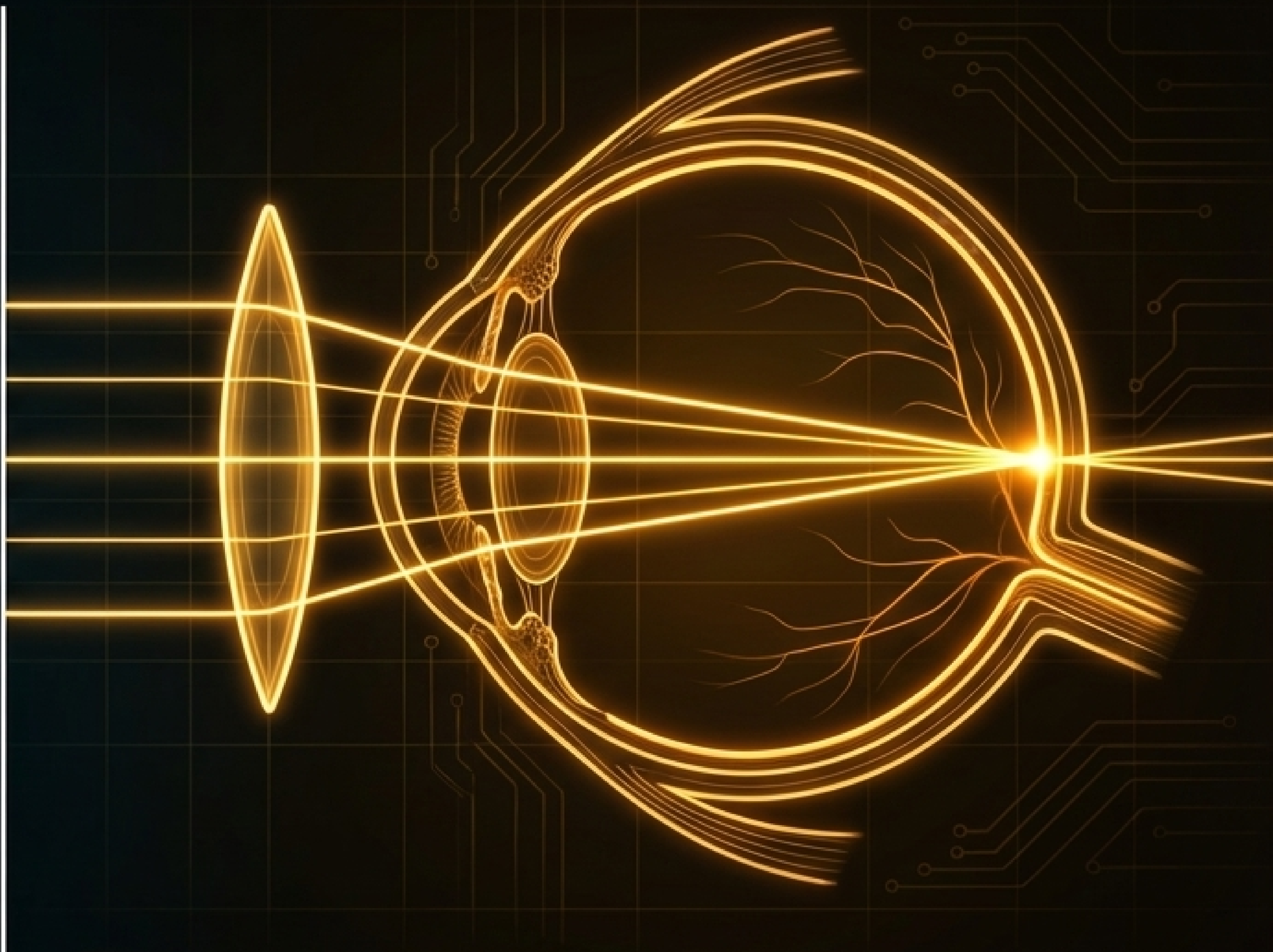
A Clinical Guide to Lensometry, Refraction, and Optical Measurement



The Physics of Correcting Refractive Errors



Minus Lenses: Diverge light to compensate for a longer eye.

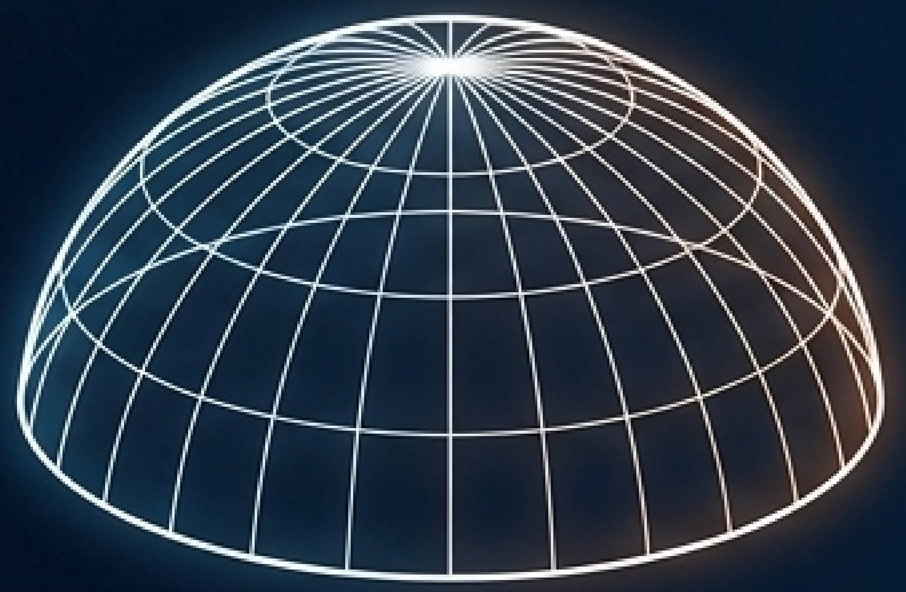


Plus Lenses: Converge light to compensate for a shorter eye (or flat cornea).

NotebookLM

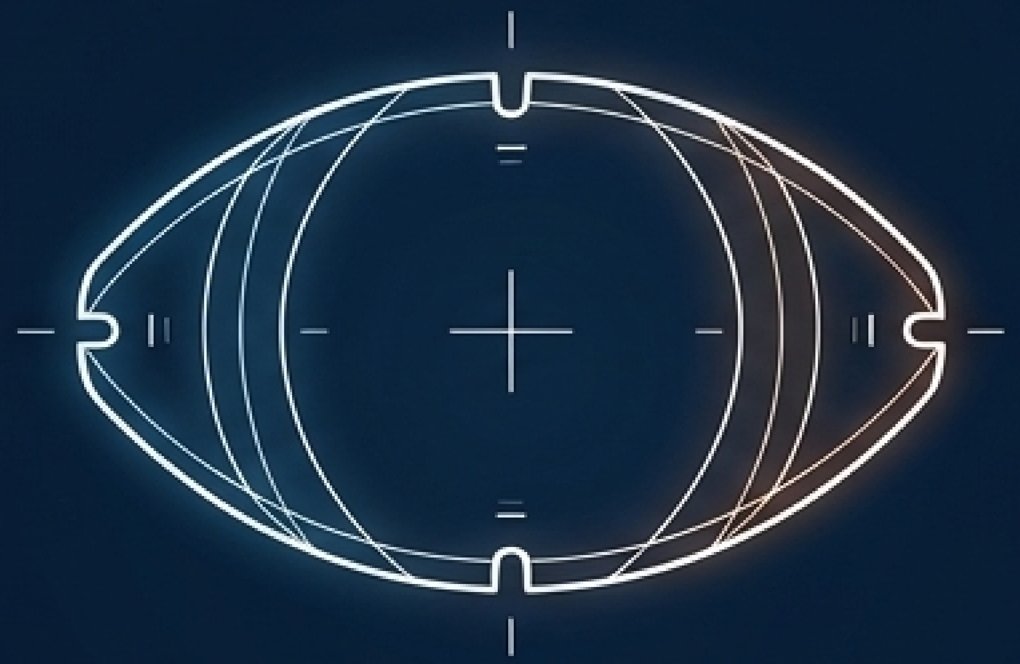
Categorizing Primary Lens Properties

Sphere



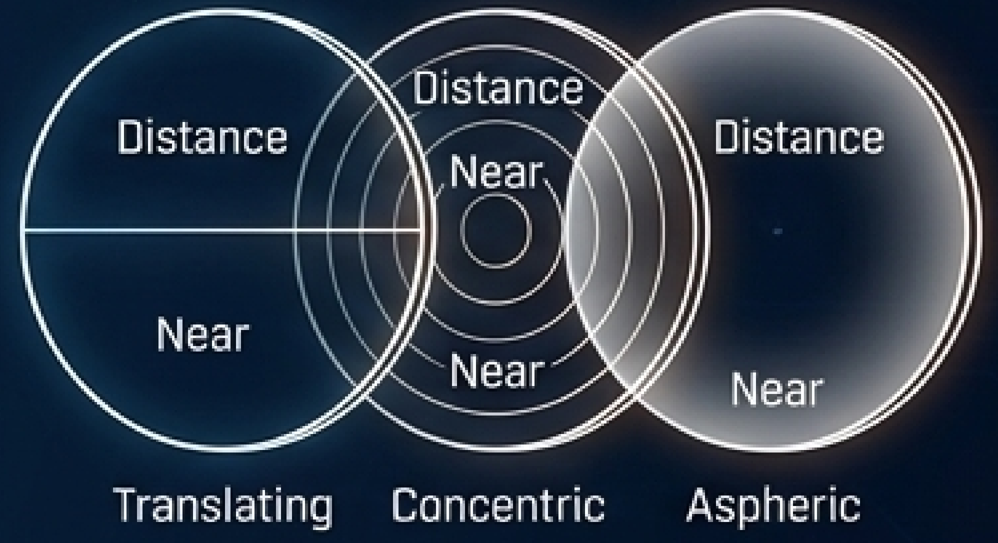
Corrects Myopia or Hyperopia. Carries equal refractive power in all meridians.

Toric



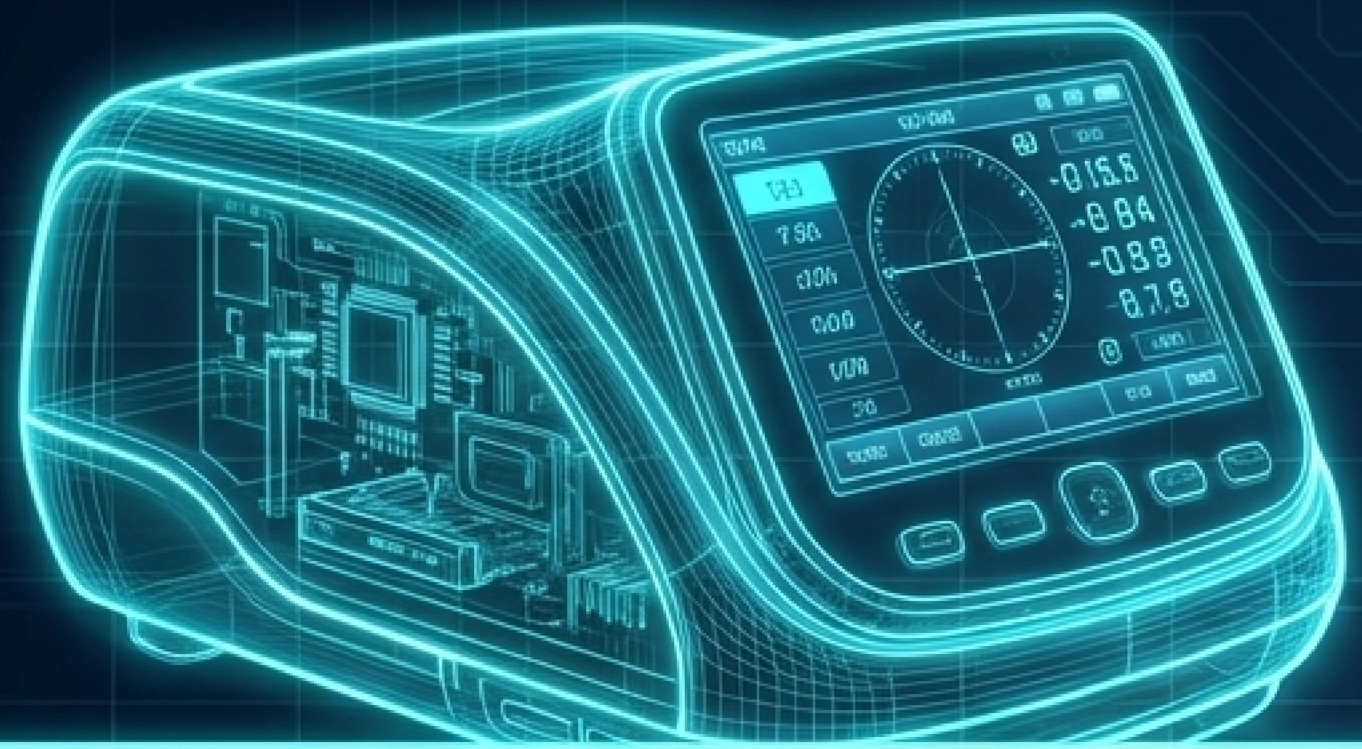
Corrects Astigmatism (where light focuses on the retina in one plane, but not another). Contains multiple focal planes and requires exact axis alignment.

Multifocal



Corrects Presbyopia (the natural loss of lens accommodation). Houses multiple focal points for distance and near vision.

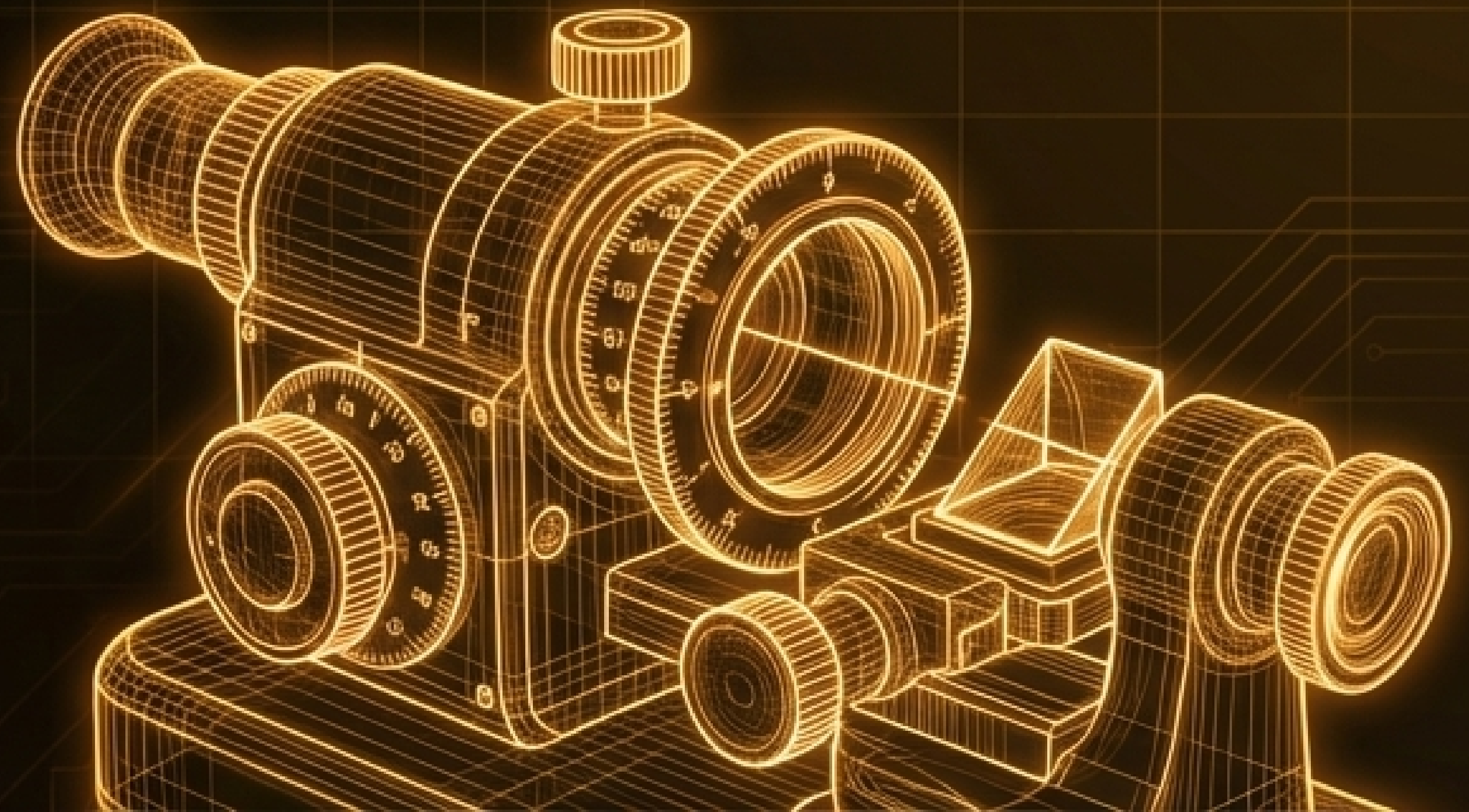
Instruments of Precision



AUTOMATED LENSOMETER

Advantage: High efficiency, prints exact Rx, ideal for high-volume clinic flow.

Limitation: Can struggle with heavily scratched lenses or complex prism measurements.



MANUAL LENSOMETER

Advantage: The clinical gold standard. Required mastery for COT and COMT JCAHPO certifications.

Limitation: Essential for troubleshooting, verifying ground-in prism, calculating slab-off, and precise progressive mark reading.

Anatomy of the Manual Lensometer

1 The Eyepiece:

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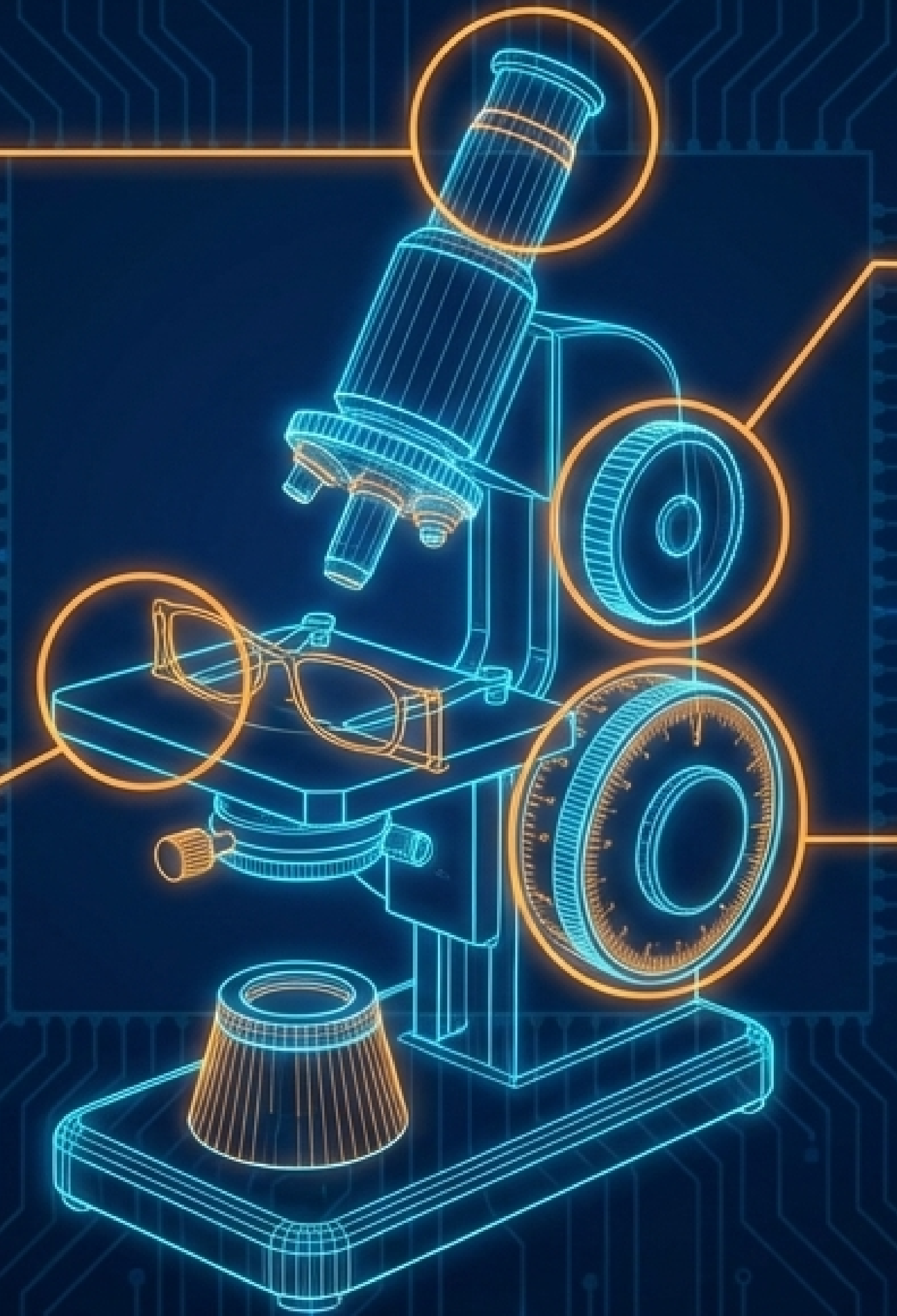

2 The Axis Wheel:

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4 The Lens Stop & Table:

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3 The Power Drum:

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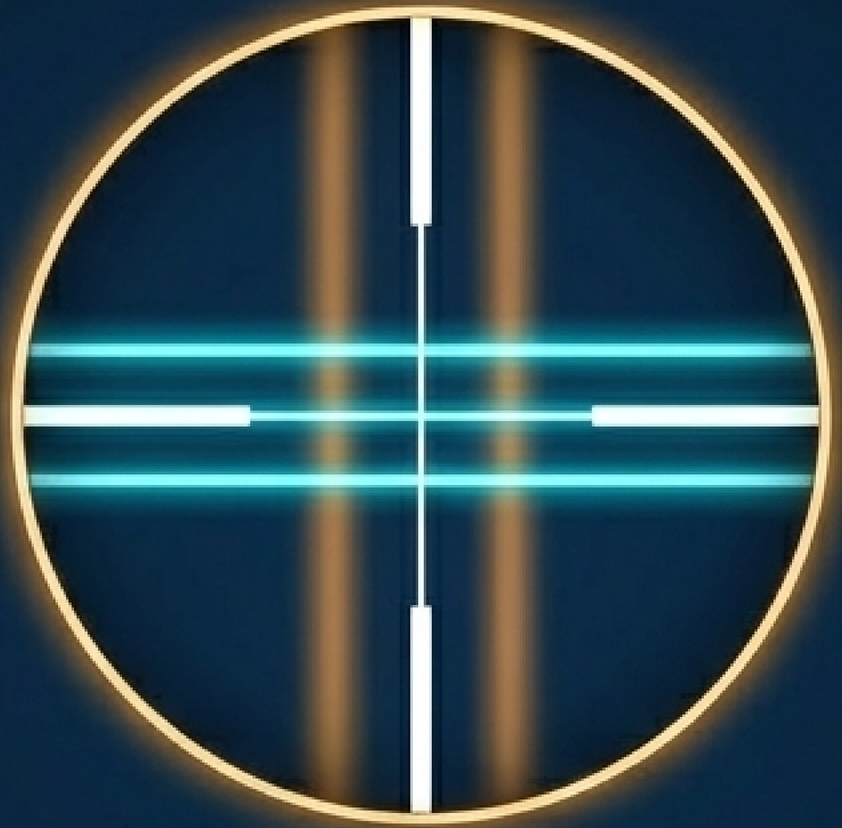
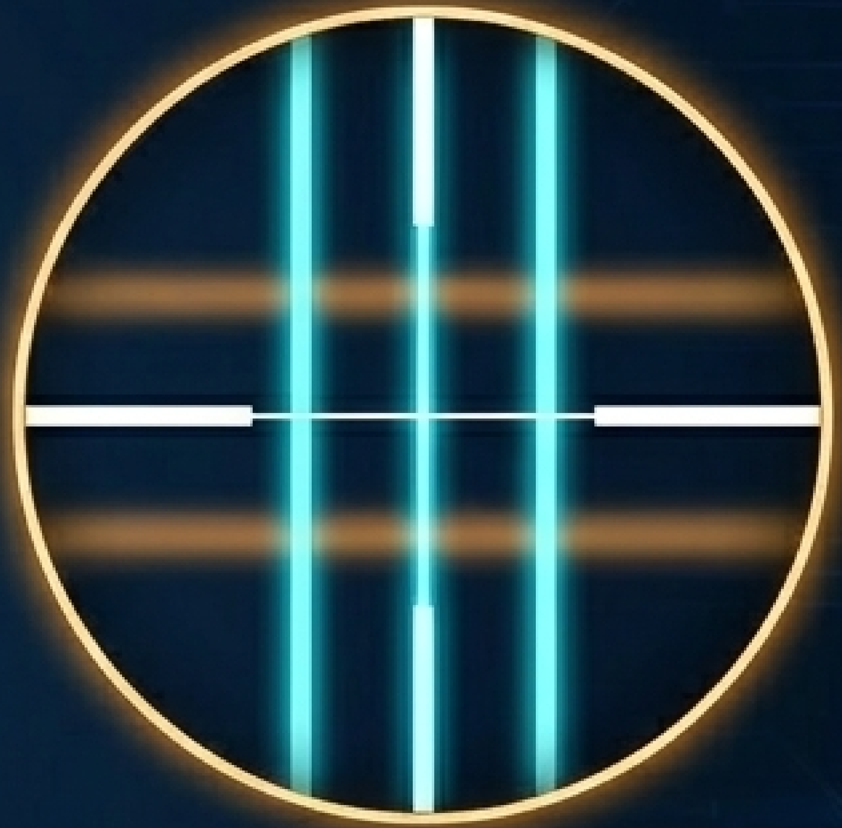
Decoding the Reticle Target

Step 1: Unneutralized → **Step 2:** Target 1 (Sphere) → **Step 3:** Target 2 (Cylinder)

Target is a blurry array of lines.



The three thin lines come into sharp, glowing cyan focus. The power drum reading at this exact moment dictates the sphere power.

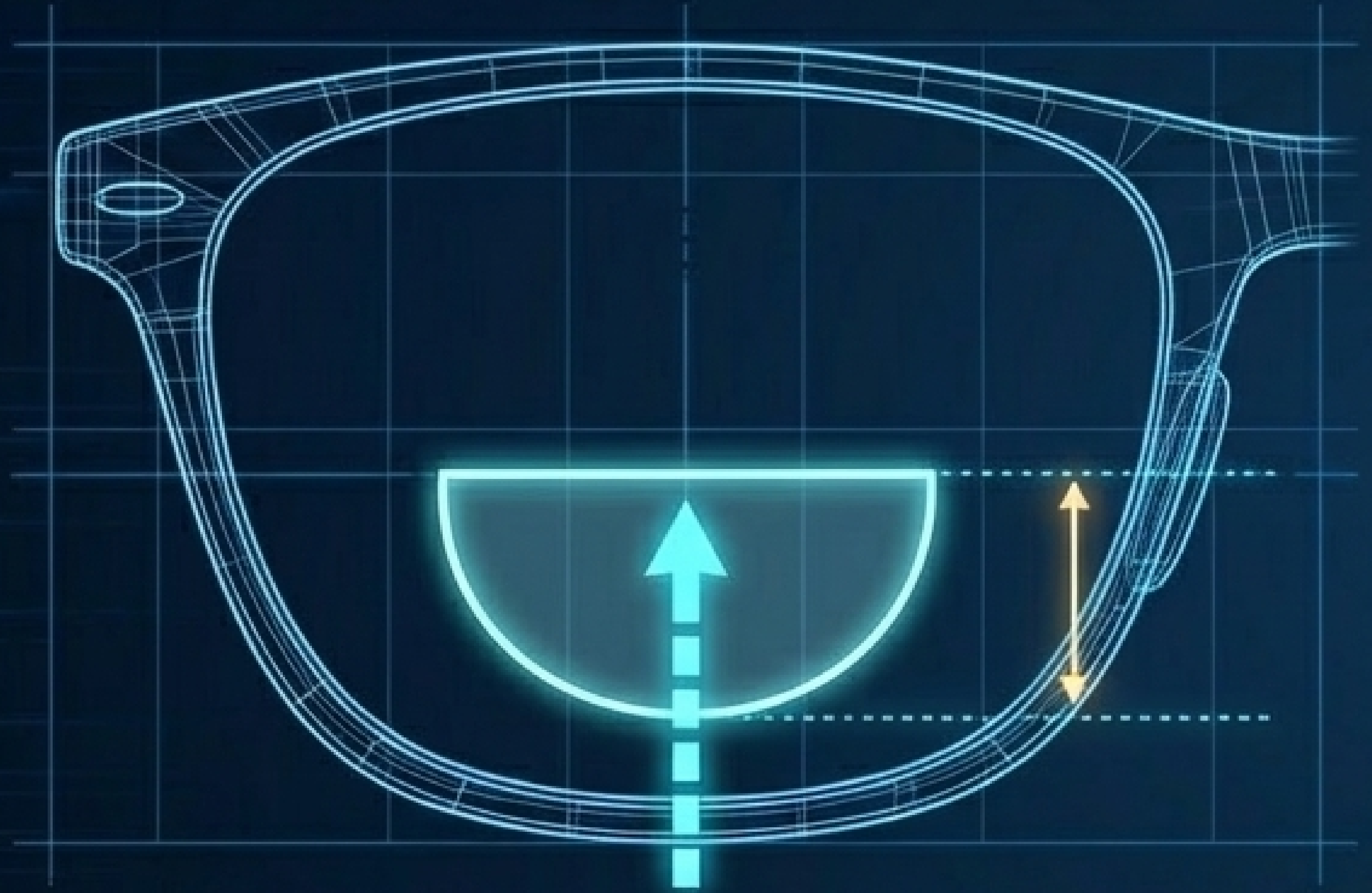


The three thick lines come into sharp focus. The dioptric difference between Target 1 and Target 2 dictates the cylinder power.

CLINICAL RULE: Always neutralize Target 1 (thin lines) before Target 2 (thick lines) when working in minus cylinder format.

Calculating the Presbyopic Add

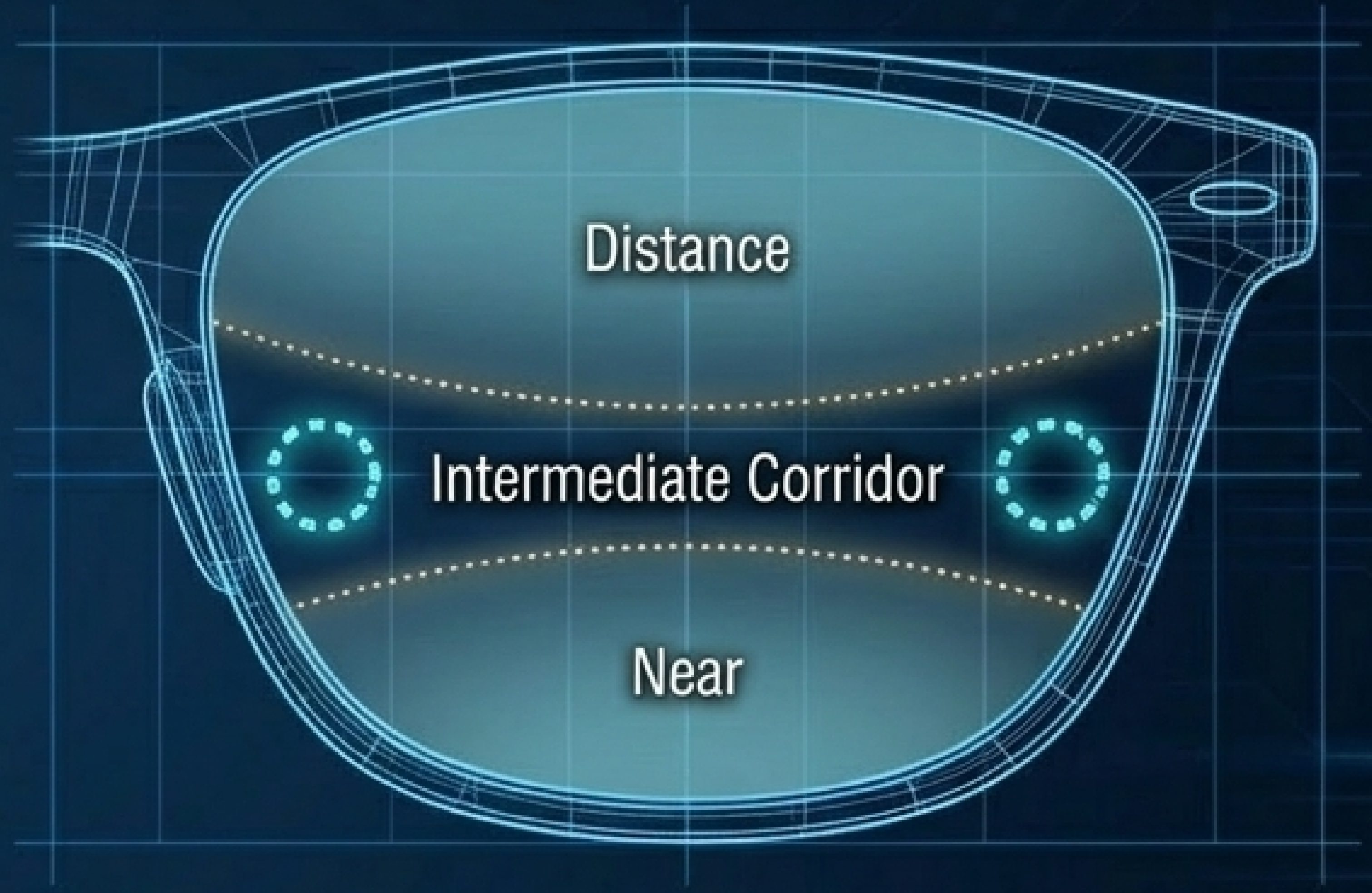
Translating (Lined Bifocal)



Process: Measure the primary distance power first. Move the lens table *up* to position the near segment over the lens stop. Read the new power.

Formula: The dioptric difference between distance and near = The Add.

Aspheric (Progressive Addition Lens)



Process: Locate the temporal circular engraving to find the pre-stamped Add power, or utilize the automated lensometer's dedicated PAL mode.

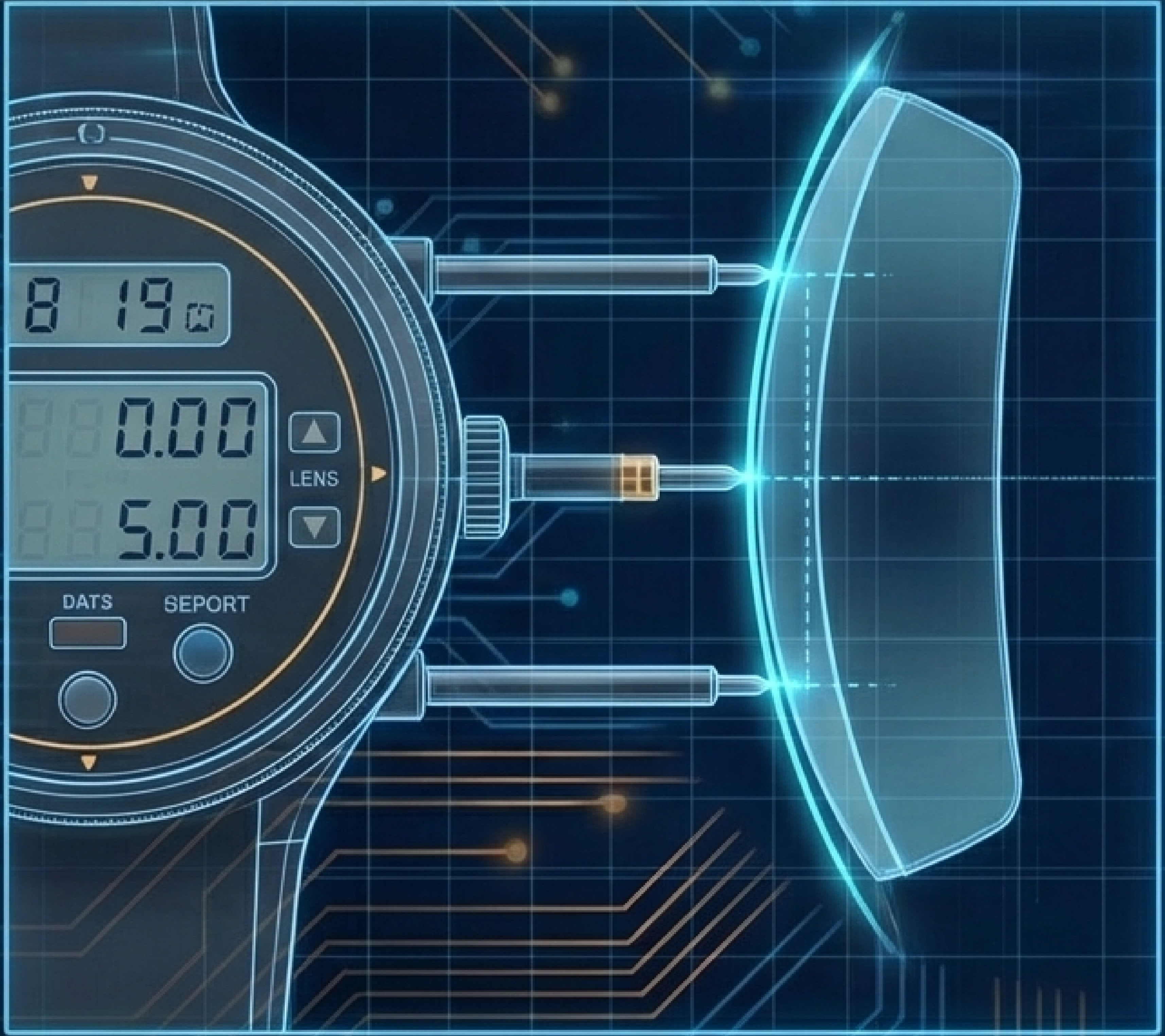
Identifying Prism Displacement

Each concentric ring represents exactly 1 diopter of prism.

Diagnostic: In this specific visual example, the target displacement toward the top right indicates Base Up and Base Out prism.

Prism is simply the optical center moving away from the physical center.

Determining Refractive Index with the Lens Clock



Step 1: Read total power on lensometer (e.g., −5.00D).

↓

Step 2: Measure Front Curve (+) and Back Curve (-) with the clock. Add them together.

↓

Step 3: Cycle the clock's index setting (1.49, 1.53, 1.56, 1.59, 1.61, 1.67, 1.71, 1.74) until the calculated curve matches the lensometer power.

Example: At index 1.61, Front reads +0.50D and Back reads -5.50D. Total = -5.00D. The material index is 1.61.

The Technician's Lensometry Workflow

1. Focus

Adjust the eyepiece to your own visual acuity.

2. Position

Place spectacles flush on the table, always beginning with OD (right eye).

3. Neutralize

Locate the sphere (thin lines) by turning the power wheel away from you.
Tip: If thick cylinder lines appear first, rotate the axis 80° (one quarter turn) and retry for spherical lines. Document that number. Then, locate and document the cylinder (thick lines) and axis.

Power Wheel Readings:
Black/Green numbers are plus (s) powers.
Red numbers are minus (-) powers.

4. Evaluate

Check for optical center displacement (prism) or multifocal add segments.

5. Document

Accurately record OD/OS, sphere, cylinder, axis, and add in the EMR.

Mastery of this workflow ensures clinical precision from the patient to the final prescription.